



Research article

Genome-wide identification and expression analysis of the metacaspase gene family in *Hevea brasiliensis*Hui Liu^a, Zhi Deng^a, Jiangshu Chen^b, Sen Wang^c, Lili Hao^c, Dejun Li^{a,*}^a Key Laboratory of Biology and Genetic Resources of Rubber Tree, Ministry of Agriculture, Rubber Research Institute, Chinese Academy of Tropical Agricultural Sciences, Danzhou 571737, China^b College of Agriculture, Hainan University, Haikou 570228, China^c CAS Key Laboratory of Genome Sciences and Information, Beijing Institute of Genomics, Chinese Academy of Sciences, Beijing 100101, China

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ABSTRACT

Metacaspases, a family of cysteine proteases, have been suggested to play important roles in programmed cell death (PCD) during plant development and stress responses. To date, no systematic characterization of this gene family has been reported in rubber tree (*Hevea brasiliensis*). In the present study, nine metacaspase genes, designated as *HbMC1* to *HbMC9*, were identified from whole-genome sequence of rubber tree. Multiple sequence alignment and phylogenetic analyses suggested that these genes were divided into two types: type I (*HbMC1*–*HbMC7*) and type II (*HbMC8* and *HbMC9*). Gene structure analysis demonstrated that type I and type II *HbMCs* separately contained four and two introns, indicating the conserved exon–intron organization of *HbMCs*. Quantitative real-time PCR analysis revealed that *HbMCs* showed distinct expression patterns in different tissues, suggesting the functional diversity of *HbMCs* in various tissues during development. Most of the *HbMCs* were regulated by drought, cold, and salt stress, implying their possible functions in regulating abiotic stress-induced cell death. Of the nine *HbMCs*, *HbMC1*, *HbMC2*, *HbMC5*, and *HbMC8* displayed a significantly higher relative transcript accumulation in barks of tapping panel dryness (TPD) trees compared with healthy trees. In addition, the four genes were up-regulated by ethephon (ET) and methyl jasmonate (MeJA), indicating their potential involvement in TPD resulting from ET- or JA-induced PCD. In summary, this work provides valuable information for further functional characterization of *HbMC* genes in rubber tree.

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1. Introduction

Programmed cell death (PCD) is a conserved and genetically controlled cell death process. In plants, PCD includes two broad categories, developmentally regulated PCD and environmentally induced PCD (Gunawardena, 2008). Developmentally regulated PCD covers a wide range of tissues and organs, such as leaf, xylem, embryo, etc. (Bollhoner et al., 2013; Huang et al., 2014; Wertman

et al., 2012), and it is initiated by the internal factors and occurs at a predictable time and location (Gunawardena, 2008). In contrast, environmentally induced PCD is triggered by external biotic or abiotic signals, such as pathogen, heat shock, and water stress (Duan et al., 2010; Kim et al., 2013; Li et al., 2012; Olvera-Carrillo et al., 2015).

PCD is essential for plant development and survival against pathogen invasion and environmental stresses. Despite the importance of PCD in plants, the molecular mechanisms involved in this process are largely unclear. However, in animals, the molecular mechanisms of PCD have been well elucidated by studying the model system *Caenorhabditis elegans* (Lord and Gunawardena, 2012). In animal cells, caspases (cysteine aspartic-specific proteases) play central role in signaling and executing PCD (Grutter, 2000). However, no orthologous caspases have been identified in plants. The only plant gene family closely resembling caspases is the metacaspase family (Uren et al., 2000). Although metacaspases have similar morphology and secondary structure as caspases, they

Abbreviations: EST, expressed sequence tag; ET, ethephon; LSD, lesion-simulating disease; MC, metacaspase; MeJA, methyl jasmonate; ORF, open reading frame; PCD, programmed cell death; qRT-PCR, quantitative real-time PCR; TPD, tapping panel dryness; TSA, transcriptome shotgun assembly.

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cannot be defined as caspases, since metacaspases do not have aspartate-specific proteolytic activity (Tsiatsiani et al., 2011). Plant metacaspases were divided into type I and type II based on their sequence and structure similarity (Tsiatsiani et al., 2011; Fagundes et al., 2015). Both type I and II metacaspases have a putative conserved caspase-like catalytic domain composed of p10 and p20 subunits, which contain conserved catalytic histidine/cysteine (His/Cys) dyad (Fagundes et al., 2015; Vercammen et al., 2007). The catalytic histidine lies in the (H/Y)(Y/F)SGHG sequence and the catalytic cysteine in the active-site, pentapeptide D(A/S)C(H/Y)S sequence (Fagundes et al., 2015; Zhang et al., 2013). Besides, type I metacaspases could present or not present a prodomain rich in proline, include a zinc finger motif in the N-terminus region. Whereas type II metacaspases lack the prodomain and the zinc finger motif, but harbor a longer linker region than that found in type I metacaspases, which connects the p10 and p20 subunits (Fagundes et al., 2015).

Metacaspases play important roles in plant PCD (Fagundes et al., 2015; Lam and Zhang, 2012). Several metacaspase genes have been demonstrated to be essential for different types of PCD in plants. In *Arabidopsis*, there are three type I (*AtMC1*–*AtMC3*, also known as *AtMCP1a*–*AtMCP1c*) and six type II (*AtMC4*–*AtMC9*, also known as *AtMCP2a*–*AtMCP2f*) metacaspase genes (Tsiatsiani et al., 2011). Among the type I metacaspases, *AtMC1* and *AtMC2* antagonistically control hypersensitive response-associated cell death in *Arabidopsis*. *AtMC1* is a positive regulator of cell death, whereas *AtMC2* negatively regulates cell death (Coll et al., 2010). Among the type II metacaspases, *AtMC4* plays a positive regulatory role in biotic and abiotic stress-induced PCD (Watanabe and Lam, 2011), and *AtMC8* is required for cell death triggered by UVC and H_2O_2 (He et al., 2008). Additionally, *AtMC9* is essential for efficient progression of autolysis during vessel cell death (Bollhoner et al., 2013; Tsiatsiani et al., 2013). In wheat, the metacaspase gene *TaMCA4* functions in PCD induced by the fungal pathogen *Puccinia striiformis* f. sp. *tritici* (Wang et al., 2012). Moreover, the pepper metacaspase gene *Camc9* plays a role as a positive regulator of pathogen-induced cell death via the regulation of reactive oxygen species production and defence-related gene expression in plants (Kim et al., 2013). In Norway spruce (*Picea abies*), the metacaspase gene *mcll-Pa* is required for both progression of vacuolar cell death and suppression of necrosis (Minina et al., 2013). These results indicate that metacaspases are essential for cell death regulation in plants.

The metacaspase gene family has been systematically investigated by genome-wide scans in Viridiplantae (Fagundes et al., 2015), *Arabidopsis* (Kwon and Hwang, 2013; Tsiatsiani et al., 2011), grape (Zhang et al., 2013), and rice (Huang et al., 2015; Wang and Zhang, 2014). In addition, several studies on metacaspases have been reported in maize (Ahmad et al., 2012), tomato (Hoeberichts et al., 2003), pepper (Kim et al., 2013), and wheat (Wang et al., 2012). However, to date, no metacaspase gene has been reported in rubber tree (*Hevea brasiliensis*).

Rubber tree is a perennial plant in the *Euphorbiaceae* family and is the sole commercial source of natural rubber because of its high production and rubber quality. Tapping panel dryness (TPD) accounts for 10–40% annual rubber production losses (Gébelin et al., 2015), therefore it is one of the most serious threats to natural rubber production. The TPD syndrome is characterized by the partial or complete cessation of latex flow upon tapping (Venkatachalam et al., 2007). Previous studies suggested that PCD in bark cells possibly play a role in TPD occurrence (Chen et al., 2003; Li et al., 2010; Putranto et al., 2015; Venkatachalam et al., 2007). Metacaspases, as important regulators of PCD, may be associated with TPD. The completion of rubber tree genome sequence has made it possible to identify and characterize the metacaspase family genes at a genome-wide level. In the present

study, we identified nine rubber tree metacaspase genes (*HbMC1*–*HbMC9*) and analyzed their gene structure, phylogenetic relationship, expression profiles in various tissues, and response to different types of abiotic stress and hormone treatments. Our results lay the foundation for future functional characterization of *HbMC* genes in rubber tree.

2. Materials and methods

2.1. Plant materials and treatments

Rubber tree clone, Reyan 7-33-97, was cultivated under normal field conditions at the experimental farm of Chinese Academy of Tropical Agricultural Sciences in Danzhou, Hainan, China. The fresh tissues or organs including leaf, stem tip, latex, bark, female flower, and male flower were sampled from 20-year-old rubber trees during spring bloom period. Roots were collected from one-year-old tissue culture seedlings of Reyan 7-33-97. Latex and barks were collected from healthy and TPD rubber trees selected according to Li et al. (2010). Three biological replicates were sampled for each tissue, and each replicate was equally harvested from five trees. Samples were immediately frozen in liquid nitrogen and then stored at -80°C for RNA isolation.

The tissue culture seedlings of Reyan 7-33-97 were used for cold, drought, and salt stress treatments. Cold stress treatment was performed by transferring seedlings to a growth chamber at 8°C . For drought and salt stress treatments, the seedlings were washed thoroughly with tap water to eliminate substrates, and then transferred into solutions supplemented with 20% (W/V) PEG-6000 (polyethylene glycol 6000) or 1 M NaCl, respectively. Each treatment had three replicates, and each replicate contained three seedlings. Leaf samples were collected at 0, 3, 24, and 48 h after treatments, and then immediately frozen into liquid nitrogen and stored at -80°C for RNA extraction.

The seven-year-old virgin trees were used for ethephon (ET), methyl jasmonate (MeJA), and wounding treatments. ET and MeJA treatments were carried out according to the methods of Hao and Wu (2000). Latex was harvested at 0, 4, 8, 24, and 48 h after treatments. The wounding treatment was performed as described by Tang et al. (2010). Latex was harvested at 0, 6, 24, and 48 h after treatment. Each treatment had three replicates, and each replicate contained three trees. The first few drops of latex containing the debris were discarded, and then the latex from the treated and control rubber tree was allowed to drop directly into liquid nitrogen in an ice kettle for total RNA extraction.

2.2. RNA isolation and first-strand cDNA synthesis

Total RNA was isolated from the collected samples according to Xu's method (Xu et al., 2010), and then treated with RQ1 RNase-free DNase (Promega, USA) to remove genomic DNA contamination. The quality and quantity of the extracted RNA were checked by agarose gel electrophoresis and measured by a spectrophotometer (Thermo Scientific NanoDrop 2000, USA). First strand cDNA was synthesized with RevertAid™ First Strand cDNA Synthesis Kit (Thermo Scientific, USA) according to the manufacturer's instruction.

2.3. Identification and isolation of metacaspase genes in *Hevea brasiliensis*

The nine full-length cDNA sequences of *Arabidopsis thaliana* metacaspase genes (*AtMC1*–*AtMC9*) were obtained from TAIR (<http://www.arabidopsis.org/>) as reported in previous study (Tsiatsiani et al., 2011). The cDNA sequences of these genes were used as queries to search against the Transcriptome Shotgun

Assembly (TSA) and Expressed Sequence Tags (EST) of *Hevea brasiliensis* at NCBI (<http://www.ncbi.nlm.nih.gov/>), and the rubber tree genome sequenced by Centre for Chemical Biology, Universiti Sains Malaysia (<http://bioinfo.ccbusm.edu.my/cgi-bin/gb2/gbrowse/Rubber/>) (Rahman et al., 2013) or by Rubber Research Institute, Chinese Academy of Tropical Agricultural Sciences (unpublished data). Redundant sequences were removed after similarity comparison. The open reading frames (ORFs) of candidate mRNA or genome DNA sequences were determined by NCBI ORF Finder (<http://www.ncbi.nlm.nih.gov/gorf/gorf.html>) and Softberry (<http://linux1.softberry.com/>). All identified HbMCs were further validated by conserved domain searching using CDD (<http://www.ncbi.nlm.nih.gov/Structure/cdd/wrpsb.cgi>) and InterProScan (<http://www.ebi.ac.uk/interpro/scan.html>) to confirm the presence of the caspase-like domain.

Based on the predicted sequences, gene-specific primers used to amplify the corresponding full length coding cDNA sequences of HbMCs were designed by Primer3.0 (<http://primer3.ut.ee/>). The primer pairs for all HbMC genes are listed in Table S1. RT-PCR was performed using Pyrobest™ DNA polymerase (TaKaRa, Japan) with the mixture of cDNA from various tissues as template. The PCR products were cloned into the pMD18-T Vector (TaKaRa, Japan) and then transformed into *E.coli* competent cells DH5 α . Finally, the products were sequenced after screening identification of bacterial colonies by PCR. At least three clones per gene were randomly picked and sequenced. The cDNA sequences of HbMCs were determined using alignment analysis with their corresponding sequences obtained from bioinformatic analysis.

2.4. Protein properties and gene structure analysis of HbMCs

The theoretical molecular weight (Mw) and isoelectric point (pI) of HbMC proteins were predicted by the ExPASy's Compute pI/Mw tool (http://web.expasy.org/compute_pi/). The genomic DNA sequences of HbMCs were obtained by the BLASTN search of the rubber tree genome database described above using the cDNA sequences as queries. Exon-intron structures of HbMCs were identified with coding sequence alignments to corresponding genomic sequences using FGENESH-C (http://linux1.softberry.com/berry.phtml?topic=fgenes_c&group=programs&subgroup=gfs).

2.5. Multiple sequence alignments and phylogenetic analysis

Amino acid sequence identity of HbMC proteins was calculated by using Clustal Omega (<http://www.ebi.ac.uk/Tools/msa/clustalo/>). The metacaspase protein and cDNA sequences of *Arabidopsis*, rice, and grape were obtained from TAIR (<http://www.arabidopsis.org/>), MSU Rice Genome Annotation Project, and GenBank according to previous studies (Tsiatsiani et al., 2011; Wang and Zhang, 2014; Zhang et al., 2013). Multiple alignments of HbMC and AtMC proteins were carried out using BioEdit software. The Bayesian phylogenetic tree was constructed using BEAST v1.8.3 (Drummond and Rambaut, 2007), which was performed as described by Cabreira et al. (2015). *Mus musculus* caspase gene *Casp1* (NM_009807.2) was used as an outgroup to root the tree.

2.6. Quantitative real-time PCR (qRT-PCR) analysis

qRT-PCR was performed with SYBR® Premix Ex Taq™ (TaKaRa, Japan) using the CFX96™ Real-Time System (Bio-Rad, USA), according to the suppliers' manuals. The thermal cycle was as follows: 95 °C for 1 min, followed by 40 cycles of 95 °C for 5 s and 60 °C for 20 s. Melting curve was routinely performed after 40 cycles to verify primer specificity. Three technical replicates were run for each biological sample. The 18S rRNA gene (GenBank accession No.:

AB268099) was used as the internal control (Tang et al., 2010). All primers were designed by Primer3 (<http://frodo.wi.mit.edu/primer3>). The primer sequences and their efficiencies are given in Table S2.

2.7. Statistical analysis

Data analysis and graphical visualization was carried out using SigmaPlot 12 software. The relative expression level was calculated using the $2^{-\Delta\Delta CT}$ method, in which CT indicates cycle threshold (Livak and Schmittgen, 2001). Data were expressed as the mean $\pm$ SD (standard deviation) of three biological replicates. Statistical analysis was performed by Tukey's test or *t*-test.

3. Results

3.1. Identification and characterization of HbMC genes in rubber tree

To identify the members of metacaspase gene family in rubber tree, the previously reported *Arabidopsis* metacaspase full-length cDNAs were used as the query sequences to search against the EST, TSA, and genome database of rubber tree with BLASTN program. The candidates were then examined by CDD and InterProScan to confirm the presence of the caspase-like domain. After removing the redundant sequences, a total of nine non-redundant metacaspase genes (designated as HbMC1–HbMC9) with complete ORFs were identified in rubber tree (Table 1).

To confirm the putative HbMCs, these complete ORF sequences of HbMCs were isolated through PCR-based approaches and sequenced. The accurate cDNA sequences of HbMCs have been deposited in GenBank with accession numbers listed in Table 1. The ORF length of HbMCs ranged from 978 bp (HbMC9) to 1254 bp (HbMC8), encoding polypeptides ranging from 325 amino acids to 417 amino acids. The corresponding molecular weights varied from 35.52 to 45.95 kDa and the predicted isoelectric points varied from 5.26 (HbMC9) to 8.60 (HbMC5) (Table 1).

Pairwise sequence comparisons were carried out to examine the degrees of sequence identities between HbMC proteins. The results are summarized in Table S3. The identities between two HbMCs ranged from 25.20% to 76.45%. The average sequence identity between two HbMCs was 40.34%. The largest identity was observed between HbMC1 and HbMC2 (76.45%). HbMC9 showed the least identities with HbMC4 and HbMC5 (25.20%).

3.2. Analysis of conserved domains and structural features of HbMCs

The conserved domain analysis indicated that all of the nine HbMC proteins contained caspase-like domain (InterPro accession No.: IPR029030), suggesting that they belonged to metacaspase gene family (Fig. S1). In addition, HbMC1–HbMC3 possessed a LSD1 (lesion-simulating disease-1)-type zinc finger domain (InterPro accession No.: IPR005735) with the consensus sequence as described by Cabreira et al. (2013), indicating that they belonged to type I with zinc finger domain metacaspases as defined by Fagundes et al. (2015).

Sequence alignment of HbMCs with AtMCs revealed the conserved motifs and structural features among metacaspases (Fig. 1). All HbMCs and AtMCs have the conserved caspase-like domain composed of p20 subunit, a linker region, and p10 subunit, containing a caspase-specific catalytic dyad of His/Cys (Uren et al., 2000). The sequence context of the catalytic His and Cys residues separately are (H/Y)(Y/F)SGHG and D(A/S)C(H/Y/N)S, which is agreed with the previously reported metacaspase catalytic

Table 1Characteristics of metacaspase family genes in *Hevea brasiliensis*.

Gene name	GenBank accession no.	ORF length (bp)	Protein			Metacaspase type
			Length (aa)	Mw (kDa)	pI	
<i>HbMC1</i>	KU188281	1104	367	40.24	6.22	I
<i>HbMC2</i>	KU188282	1086	361	39.55	6.59	I
<i>HbMC3</i>	KU188283	1164	387	42.46	5.90	I
<i>HbMC4</i>	KU188284	1164	387	43.58	6.89	I
<i>HbMC5</i>	KU188285	1065	354	39.92	8.60	I
<i>HbMC6</i>	KU188286	1086	363	40.82	8.24	I
<i>HbMC7</i>	KU188287	1026	341	38.47	6.56	I
<i>HbMC8</i>	KU188288	1254	417	45.95	5.01	II
<i>HbMC9</i>	KU188289	978	325	35.52	5.26	II

ORF, open reading frame; bp, base pair; aa, amino acids; Mw, molecular weight; pI, isoelectric point.

site sequences (Fagundes et al., 2015).

HbMC1–*HbMC7* showed structural similarity to *AtMC1*–*AtMC3* (Fig. 1). Their proteins contain a proline- or glutamine-rich N-terminal prodomain and a shorter linker region between the p20 and p10 subunits, which was in accordance with the characteristics of type I metacaspases. Thus, they belonged to type I *HbMCs*. The protein structural of *HbMC8* and *HbMC9* were similar to *AtMC4*–*AtMC9* (Fig. 1). They lack the N-terminal prodomain but harbor a longer linker region between the p20 and p10 subunits, which corresponds to the characteristics of type II metacaspases. Thus, they belonged to type II *HbMCs*.

3.3. Gene structures and phylogenetic analysis of *HbMCs*

In order to gain further insight into the structural diversity of *HbMC* genes, we performed an exon/intron analysis by aligning ORF sequences of *HbMCs* with their corresponding genomic sequences (Fig. 2). According to their predicted structures, the nine *HbMCs* could be divided into two groups. The first group *HbMCs* consists of five exons interrupted by four introns, including all the type I *HbMCs* (*HbMC1*–*HbMC7*). The other group included all the type II *HbMCs* (*HbMC8* and *HbMC9*), which contain two exons interrupted by one intron. These results suggested that *HbMCs* within the same type shared conserved exon–intron structures.

To obtain insight into the evolutionary history and phylogenetic relationships of the *HbMC* genes, a Bayesian phylogenetic tree was constructed using BEAST software on the basis of multiple sequence alignment of metacaspase family genes from *Arabidopsis*, rubber tree, rice, and grape (Fig. 3). According to the phylogenetic tree, the 32 metacaspase genes were divided into two clades (I and II). Clade I consisted of 18 type I metacaspase genes, and it was further categorized into three subclades: A, B, and C, with each subclade containing seven, seven, and four members, respectively. Clade II contained 14 type II metacaspase genes, and it was also divided into three subclades: D, E, and F, with each subclade containing four, three, and seven members, respectively. The nine *HbMCs* were distributed in four subclades: A (*HbMC1* and *HbMC2*), B (*HbMC3*–*HbMC7*), D (*HbMC9*), and F (*HbMC8*). Subclades C and E were rice-specific subclades.

3.4. Expression profiles of *HbMC* genes in different tissues of rubber tree

To investigate the tissue specificity of *HbMCs* expression, the expression patterns of *HbMCs* in various tissues (including roots, leaves, stem tips, barks, latex, male flowers, and female flowers) of rubber tree were analyzed using qRT-PCR. As shown in Fig. 4, *HbMCs* displayed different tissue expression patterns. *HbMC1*, *HbMC2*, *HbMC5*, and *HbMC8* were constitutively expressed in all

tissues tested. Neither *HbMC3* nor *HbMC9* expression was detected in barks, while *HbMC6* and *HbMC7* transcripts were not detected in both barks and latex. *HbMC4* showed no expression in latex and a significantly lower expression in barks. In contrast, some genes were highly expressed in specific tissues. For example, *HbMC8* had a significantly higher expression in latex than other tissues. *HbMC5* and *HbMC6* had a significantly higher expression in leaves than other tissues. In addition, *HbMC1*, *HbMC7* and *HbMC9* were relatively highly expressed in roots and stem tips (Fig. 4). The *HbMCs* highly expressing in specific tissues may play specific roles in the corresponding tissues.

Systematic analyses of TPD-related genes suggested that PCD might play important roles in rubber tree TPD (Gébelin et al., 2015; Li et al., 2010; Venkatachalam et al., 2007). Metacaspases, as key regulators of PCD, possibly play critical roles in TPD occurrence. Therefore, we comparatively analyzed *HbMCs* expressions between healthy and TPD rubber trees. Among the nine *HbMCs*, *HbMC3*, *HbMC4*, *HbMC6*, *HbMC7*, and *HbMC9* showed no transcript or no significant changes in latex and barks of healthy and TPD rubber trees (data not shown), indicating that they may not be involved in TPD. In contrast, *HbMC1*, *HbMC5*, and *HbMC8* showed significantly higher expression in both latex and barks of TPD trees than that of healthy trees (Fig. 5). Additionally, the relative expression level of *HbMC2* was significantly higher in barks of TPD trees than that of healthy trees (Fig. 5). These results suggested that *HbMC1*, *HbMC2*, *HbMC5*, and *HbMC8* may be involved in TPD occurrence.

3.5. Expression patterns of *HbMCs* in response to abiotic stress

Metacaspases are important regulators of PCD during stress responses in plants (Fagundes et al., 2015; Huang et al., 2015). To explore the possible involvement of *HbMCs* in response to abiotic stress, the expression patterns of *HbMCs* under cold, drought, and salt stresses were analyzed by qRT-PCR. As shown in Fig. 6, the expressions of six *HbMC* genes were regulated by cold stress, and the rest three genes, *HbMC2*, *HbMC5*, and *HbMC6*, showed no obvious transcriptional changes under cold stress. Among the six cold-regulated *HbMCs*, *HbMC1*, *HbMC7*, and *HbMC9* were significantly up-regulated at all the treated time-points. *HbMC8* showed significant up-regulation only at 24 h after treatment. Only two *HbMC* genes, *HbMC3* and *HbMC4*, exhibited down-regulated expression in response to cold stress. *HbMC4* were dramatically suppressed at 3 and 24 h, while *HbMC3* showed significant down-regulation at 24 and 48 h (Fig. 6).

Under PEG-induced drought stress, only *HbMC5* expression was not significantly changed (Fig. 7). Of the eight drought-responsive *HbMC* genes, four genes (*HbMC1*, *HbMC2*, *HbMC8*, and *HbMC9*) showed significant up-regulation at least one time-point after

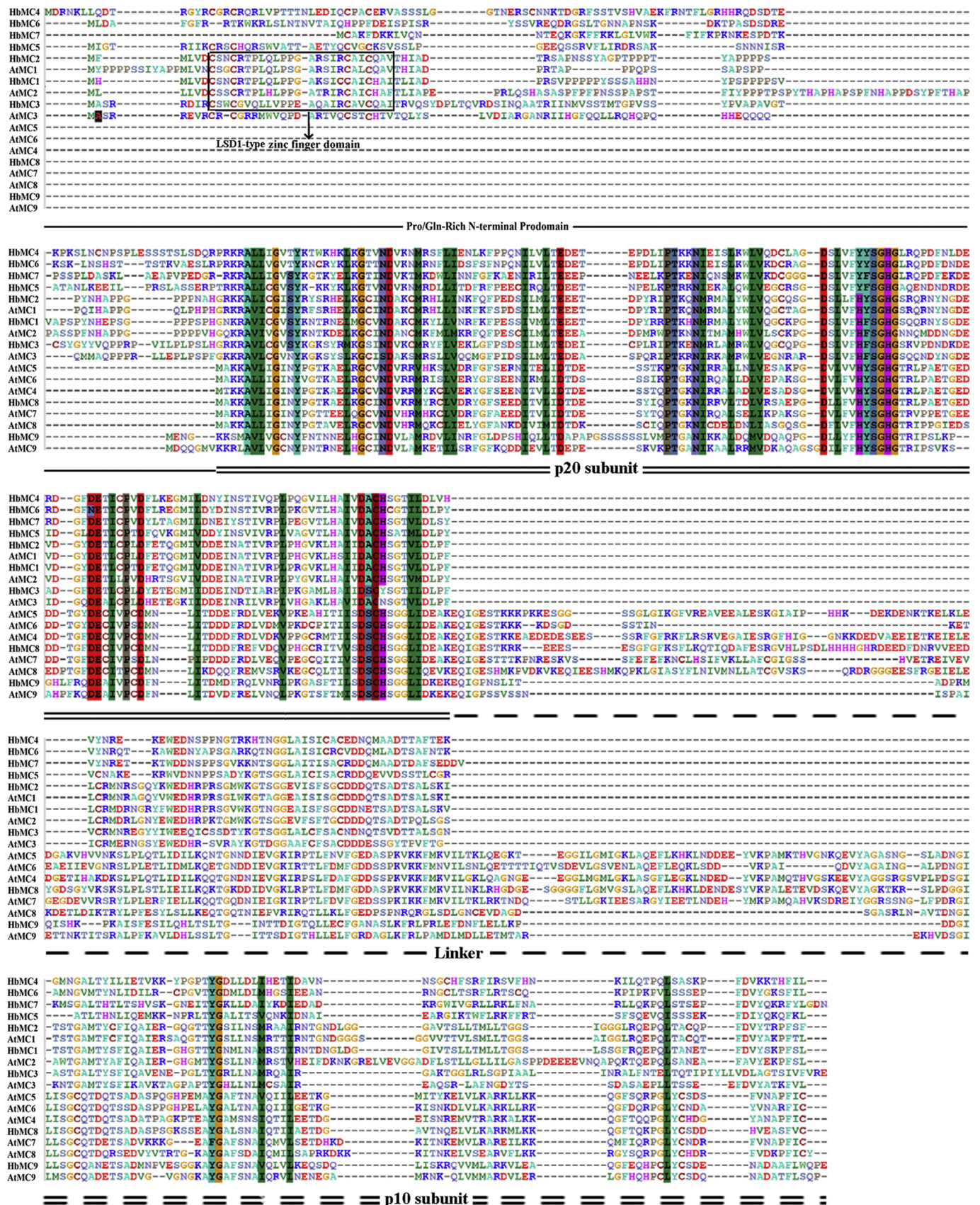


Fig. 1. Multi-sequence alignment of metacaspase proteins from rubber tree (Hb) and *Arabidopsis* (At). The rectangle indicate LSD1-type zinc finger domain. The solid line, double solid line, dotted lines, and double dotted lines indicate the Pro/Gln-Rich N-terminal Prodomain, p20 subunit, linker, and p10 subunit, respectively.

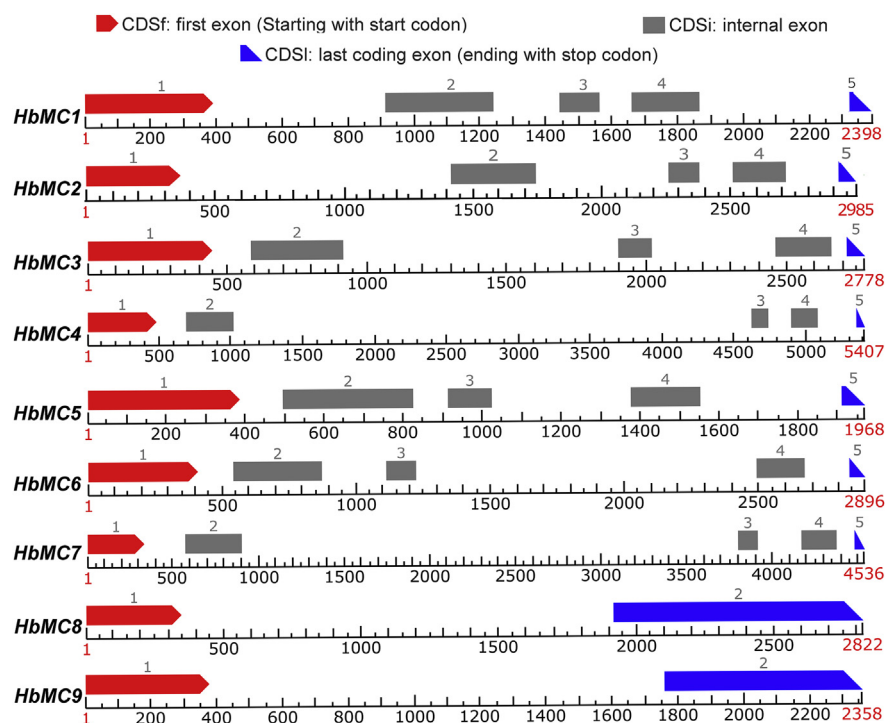


Fig. 2. Exon–intron structures of *HbMC* genes. The first exons are represented by red boxes. Internal exons are represented by grey boxes and the last exons are represented by blue boxes. Scales show the length of each gene's exons and introns in bp. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

treatment (Fig. 7). Among them, *HbMC1* and *HbMC2* displayed similar expression patterns upon drought stress. Both of them were rapidly and strongly induced by drought stress, reached the maximum level at 3 h after treatment (6.4- and 2.6-fold of the control for *HbMC1* and *HbMC2*, respectively), and then declined. In contrast, *HbMC9* transcripts exhibited continuous increases after drought stress, and resulting in 4.6-fold increase at 48 h after treatment. *HbMC8* showed significant up-regulation at 24 h, only increased 1.5-fold over that at 0 h. A total of four *HbMC* genes were down-regulated by drought stress (Fig. 7). *HbMC6* showed significant down-regulation at all the treated time-points. Both *HbMC4* and *HbMC7* exhibited down-regulated expression at 24 and 48 h after treatment. While *HbMC3* displayed significant repressed expression only at 24 h after treatment.

Under salt stress, *HbMC2*, *HbMC4*, and *HbMC7* showed similar expression patterns (Fig. 8). Their transcripts were rapidly up-regulated in response to salt stress, peaked at 3 h of treatment, and then declined. After 48 h of treatment, the transcript abundance of *HbMC2* recovered to normal level, whereas the expression levels of *HbMC4* and *HbMC7* declined to less than 0.2-fold of the control. Additionally, the expression of *HbMC3* and *HbMC6* also showed significant down-regulation at relatively later stages (24 and/or 48 h after treatment). By contrast, *HbMC1* exhibited up-regulated expression at 24 h after treatment (Fig. 8). The remaining three *HbMC* genes (*HbMC5*, *HbMC8*, and *HbMC9*) displayed no significant expression change after salt stress treatment (Fig. 8).

3.6. Expression patterns of *HbMCs* in response to wounding

Farmers harvest latex by regularly tapping the trunk bark of the rubber tree. Tapping is mechanical wounding, and latex leakage is a wounding response of rubber tree. To investigate whether mechanical wounding had any effect on *HbMCs*, the expression patterns of *HbMCs* in latex after wounding were analyzed (Fig. 9). No

HbMC4, *HbMC6*, and *HbMC7* transcripts were detected in latex after wounding treatment, which was in accordance with the tissue expression patterns of *HbMC4*, *HbMC6*, and *HbMC7*. Additionally, *HbMC2* and *HbMC9* showed no obvious transcription level change (data not shown), suggesting they may not be involved in rubber tree wounding response. The expressions of four *HbMCs* in latex were induced or repressed after wounding treatment (Fig. 9). Of these, *HbMC1* expression was significantly up-regulated in response to wounding across all time-points, whereas *HbMC3* exhibited significantly down-regulated expression in latex in response to wounding across all time-points. Besides, *HbMC5* and *HbMC8* were significantly up-regulated by wounding treatment at different stages. *HbMC5* was significantly induced, leading to approximately 3.8-fold increase at 24 h after treatment, while *HbMC8* was slightly increased by 1.6-fold at 48 h after treatment, as compared with that in control plants without wounding treatment.

3.7. Expression patterns of *HbMCs* in response to ET and MeJA

Jasmonic acid and ethylene regulate cell death under stress conditions and during development (Lam, 2004). Moreover, ethylene and jasmonic acid play critical roles in regulating latex production in rubber tree (Hao and Wu, 2000; Zhu and Zhang, 2009). Given the major roles of ethylene and jasmonic acid in regulating latex biosynthesis and cell death, we further analyzed the expression of *HbMCs* in latex in response to exogenous ET and MeJA treatments. *HbMC4*, *HbMC6*, and *HbMC7* still showed no expression in latex after ET and MeJA treatments. Additionally, there was no significant change in the expressions of *HbMC3* and *HbMC9* (data not shown). Only four *HbMCs* were regulated by ET and JA treatments (Fig. 10). After ET treatment, *HbMC1* and *HbMC5* displayed significantly up-regulated expression across all time-points, and *HbMC2* exhibited significant up-regulation only at 8 h. *HbMC8* displayed an irregular expression pattern. It was up-

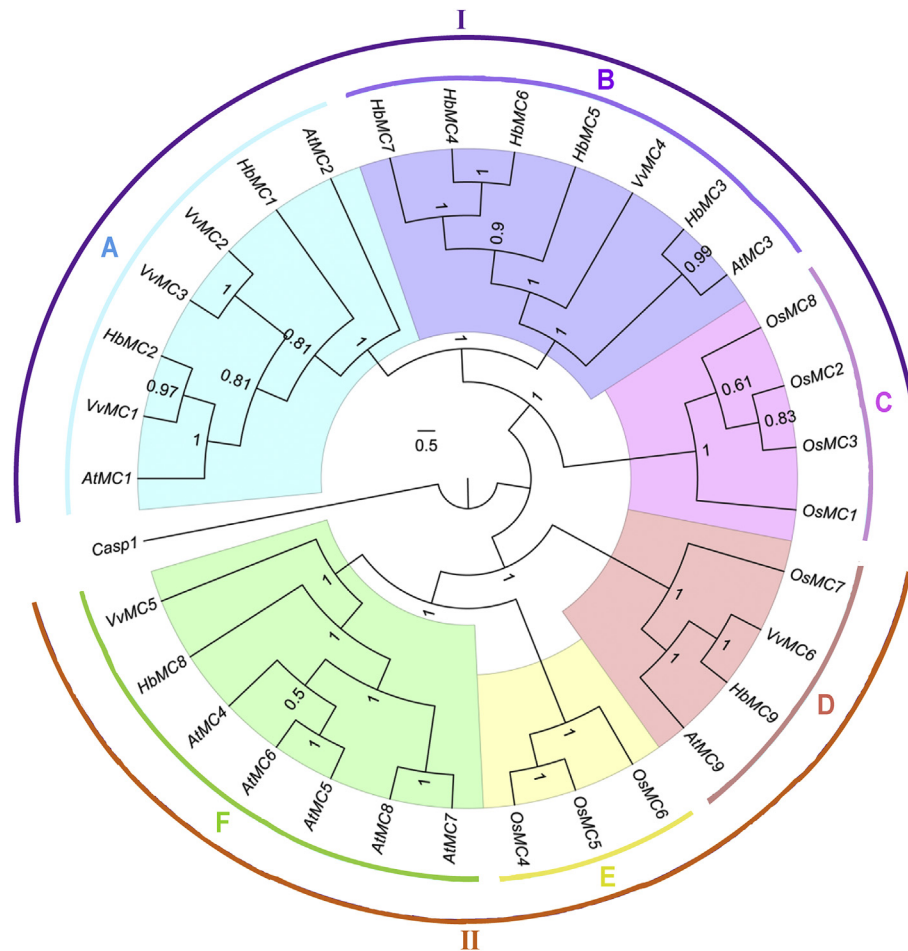


Fig. 3. Phylogenetic relationships of the metacaspase family proteins from rubber tree, *Arabidopsis*, rice and grape. The Bayesian phylogenetic tree was constructed using BEAST v1.8.3. The caspase gene *Casp1* (NM_009807.2) from *Mus musculus* was used as an outgroup to root the tree. The posterior probabilities are given for each node in the tree. The six subclades are indicated with different colors. The metacaspase family genes of *Arabidopsis* (At), rice (Os), and grape (Vv) were described according to previous studies (Tsiatsiani et al., 2011; Wang and Zhang, 2014; Zhang et al., 2013). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

regulated at 8 h, but was suppressed at 4 and 48 h. Interestingly, all of these four *HbMCs* showed the highest expression levels at 8 h, 14.5-, 2.3-, 10.6-, and 1.9-fold up-regulation for *HbMC1*, *HbMC2*, *HbMC5*, and *HbMC8*, respectively. After MeJA treatment, *HbMC1*, *HbMC2*, and *HbMC5* transcripts increased gradually, peaked at 8 h, and then recovered to normal levels after 24 h. *HbMC8* transcripts were also gradually accumulated, but it reached the maximum level at 24 h after treatment, and increased only 1.6-fold over that at 0 h.

4. Discussion

Metacaspases are cysteine proteases and widespread presence in Viridiplantae, from algae to vascular plants. The number of metacaspase genes in the genomes of different plant species varies considerably, from one in some of green algae to 20 in *Brassica rapa* Chiifu-FPsc (Fagundes et al., 2015). However, to our knowledge, the metacaspase gene family from rubber tree has not been characterized in detail. In this study, we carried out genome-scale identification and expression analysis of metacaspase gene family in rubber tree. A total of nine *HbMC* genes were identified in rubber tree genome, which is equal to the number of metacaspase genes in *Arabidopsis* (Tsiatsiani et al., 2011). Of the nine *AtMCs*, three belonged to the type I metacaspases, and the other six were type II metacaspases. However, the rubber tree possessed seven type I metacaspases and only two type II metacaspases (Figs. 1 and 3).

More recently, Fagundes et al. (2015) adopted a comparative genomic approach to identify metacaspase genes in Viridiplantae. They predicted the distribution of type I and II metacaspases in 42 plant species. The total number of type I metacaspases was 259, which is more than twice the number of type II metacaspases, suggesting that most of plant species have greater numbers of type I metacaspases relative to type II.

Gene structure analysis revealed that type I *HbMCs* had five exons interrupted by four introns, and type II *HbMCs* consisted of two exons and a single intron (Fig. 2). The same exon/intron structures were obtained from the analysis of grape *VvMCs* (Zhang et al., 2013) and *Arabidopsis* type II *AtMC4* (Watanabe and Lam, 2011). We further analyzed the gene structure of the other *AtMCs* and found that type I *AtMC1*–*AtMC3* also had the 5 exon/4 intron structures, and type II *AtMC5*–*AtMC8* also had the 2 exon/1 intron structures. However, type II *AtMC9* only had one exon. Rice type I *OsMCs* had 3–5 exons, and type II *OsMCs* had one or two exons (Wang and Zhang, 2014). These results suggested that type I metacaspases had more exon/intron numbers than type II metacaspases.

The type I metacaspases may present or not present a zinc finger domain in the N-terminus region (Fagundes et al., 2015). In the present study, zinc finger domain was found in type I metacaspases *HbMC1*–*HbMC3* but not in *HbMC4*–*HbMC7*. It has been hypothesized that the acquisition of the zinc finger domain seems to have

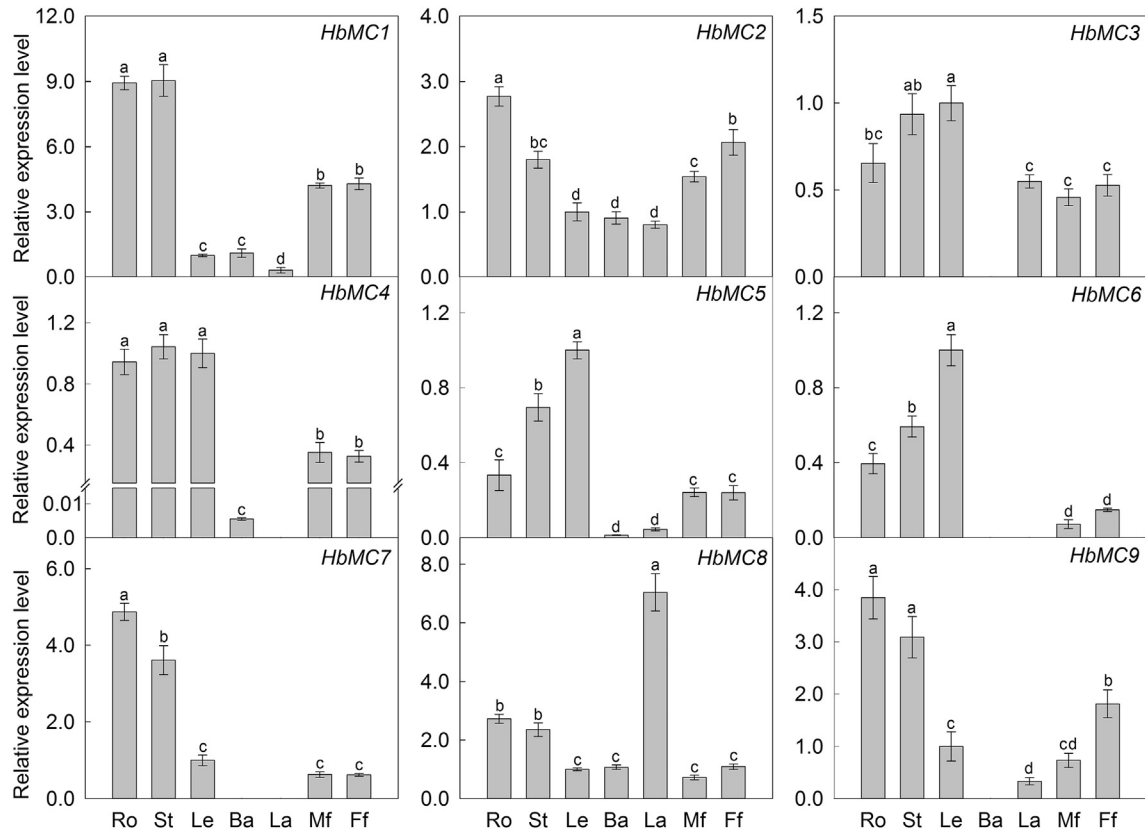


Fig. 4. Expression patterns of *HbMC* genes in different tissues of rubber tree. Relative expression levels of *HbMC* genes were determined by qRT-PCR and normalized by the *18S rRNA* gene expression. For each gene, the transcript level in leaf was used to normalize the transcript levels in other tissues. Values are means $\pm$ SD of three biological replicates. Different letters indicate significant differences among the different tissues ($P < 0.05$, Tukey's test). Ro, roots; St, stem tips; Le, leaves; Ba, Barks; La, latex; Mf, male flowers; Ff, female flowers.

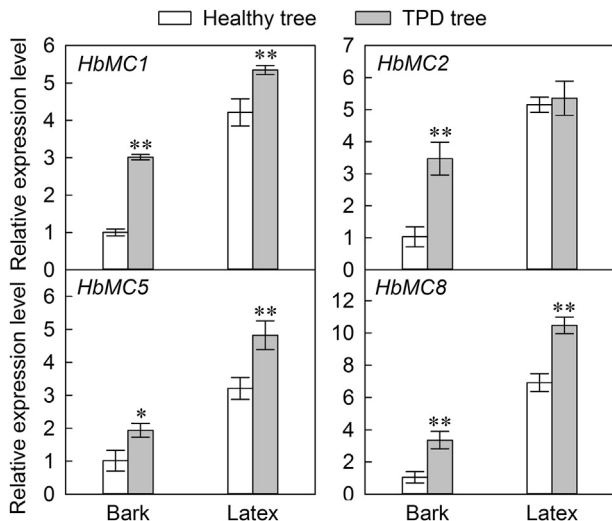


Fig. 5. Comparative analysis of expression levels of *HbMCs* in latex and barks of the healthy and TPD rubber trees. Relative expression levels of *HbMC* genes were determined by qRT-PCR and normalized by the *18S rRNA* gene expression. For each gene, the transcript level in barks of healthy tree was used to normalize the transcript level in other tissues. Values are means $\pm$ SD of three biological replicates. Asterisks indicate a significant difference (*, $P < 0.05$; **, $P < 0.01$, *t*-test) between the TPD and healthy rubber trees.

occurred later during the metacaspase gene family evolution (Fagundes et al., 2015). Thus, *HbMC4*–*HbMC7* should be originated in the early stage of *Hevea brasiliensis* evolution. It has been proved

that the LSD1-type zinc finger domains in *Arabidopsis* AtMC1 and rice OsMC1 were required for interaction with LSD proteins (Coll et al., 2010; Huang et al., 2015). Whether *HbMC1*–*HbMC3* can interact with HbLSDs and their zinc finger domains are required for the interaction still need to be examined.

Comprehensive gene expression analysis of metacaspase family genes revealed that the metacaspases had distinct expression patterns in various tissues of *Arabidopsis*, rice, and grape (Kwon and Hwang, 2013; Wang and Zhang, 2014; Zhang et al., 2013). Consistent with the aforementioned results, *HbMCs* also showed apparent differential expression patterns in various tissues. Some *HbMCs* exhibited high expression levels in specific tissues, implying that they might play specific roles required in these tissue types. For example, *HbMC1*, *HbMC7*, and *HbMC9* were highly expressed in roots and stem tips, and *HbMC6* was abundantly expressed in leaves. Six *HbMCs* were expressed in latex. Of these, *HbMC8*, also designated as latex-abundant protein (GenBank: AAD13216.1), showed a relatively higher expression level in latex than in other tissues (Fig. 4). Furthermore, the expression of *HbMC8* was regulated by ET and MeJA (Fig. 10). ET and MeJA are key signals for latex production in rubber tree (Hao and Wu, 2000; Zhu and Zhang, 2009). These results suggested that *HbMC8* may play an important role in latex biosynthesis in rubber tree.

Hevea brasiliensis is a native species of the Amazon Basin of South America, but it is widely planted in Southeast Asia, such as Thailand, Vietnam, southern China, etc. Rubber trees planting in these new areas are often subjected to abiotic stresses like low temperature, drought, and typhoons. Diverse abiotic stresses have been found to induce PCD (Petrov et al., 2015). Previous studies have pointed to metacaspase involvement in abiotic-induced PCD

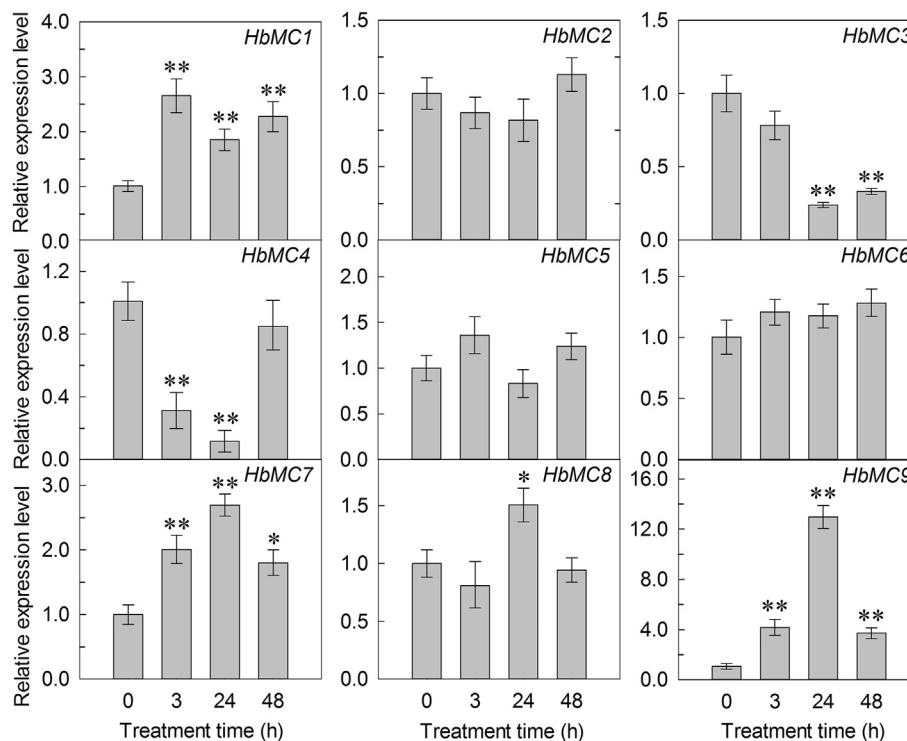


Fig. 6. Expression profiles of *HbMC* genes under cold stress. Leaves of seedlings were sampled at 0, 3, 24 and 48 h after cold (8 °C) stress treatment. Relative expression level of each gene was normalized with *18S rRNA* gene. Data are means $\pm$ SD of three biological replicates. Asterisks indicate a significant difference (*, $P < 0.05$; **, $P < 0.01$, *t*-test) compared with the corresponding control (0 h).

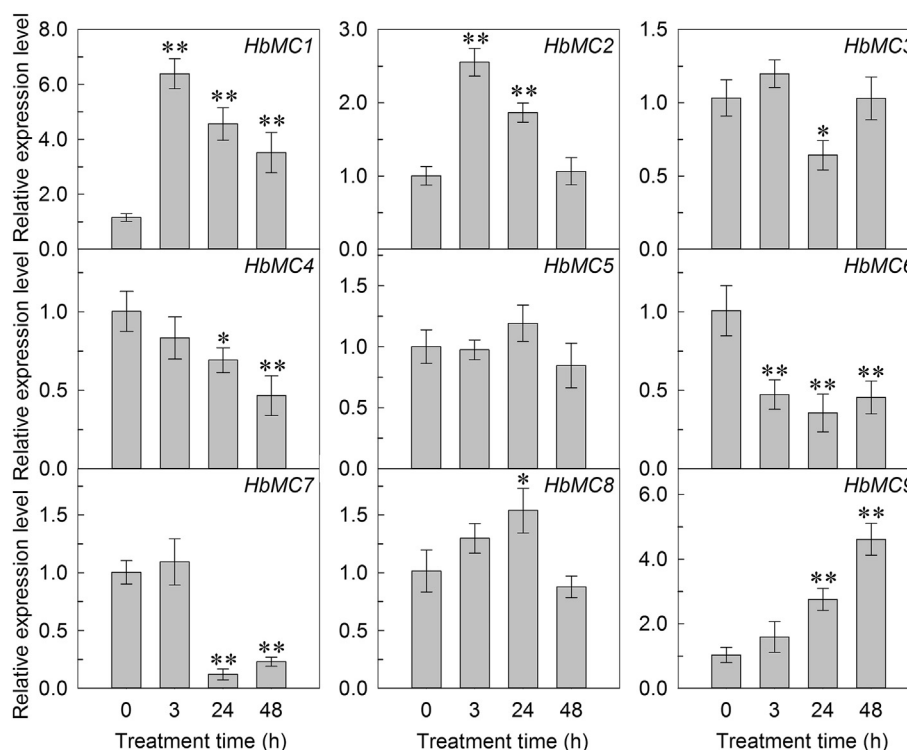


Fig. 7. Expression profiles of *HbMC* genes under drought stress. Leaves of seedlings were sampled at 0, 3, 24 and 48 h after PEG-induced drought stress treatment. Relative expression level of each gene was normalized with *18S rRNA* gene. Data are means $\pm$ SD calculated from three biological replicates. Asterisks indicate a significant difference (*, $P < 0.05$; **, $P < 0.01$, *t*-test) compared with the corresponding control (0 h).

in pants (Fagundes et al., 2015). In *silico* analysis of *cis*-elements in the promoter sequences of grapevine and rice metacaspase family

genes indicated that all of these metacaspase gene promoters contained *cis*-elements related to stress responses (Huang et al.,

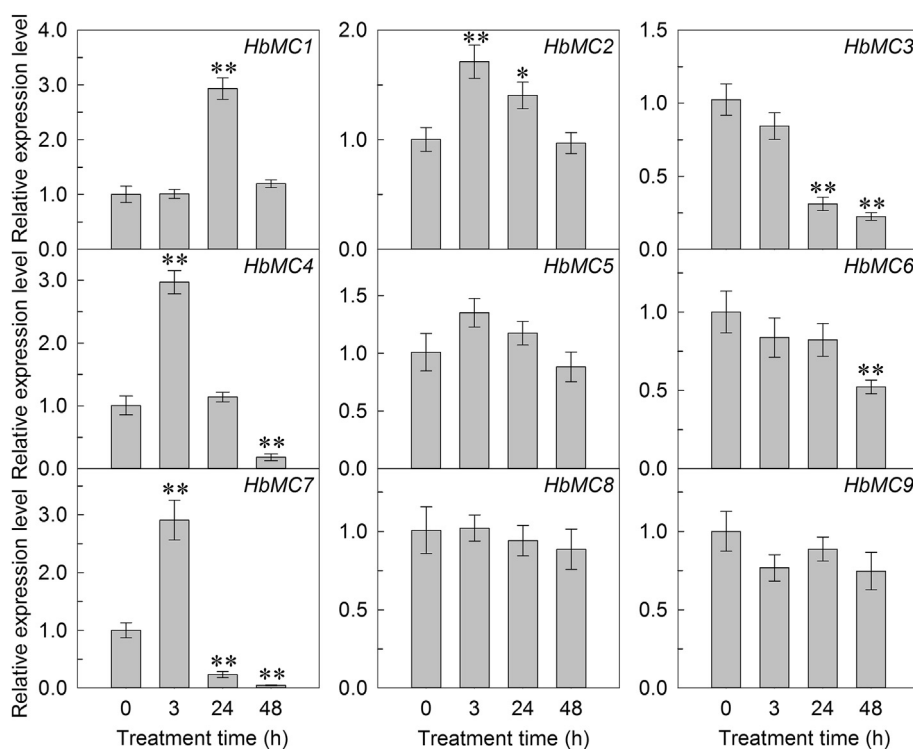


Fig. 8. Expression profiles of *HbMC* genes under salt stress. Leaves of seedlings were sampled at 0, 3, 24 and 48 h after salt (1 M NaCl) stress treatment. Relative expression level of each gene was normalized with *18S rRNA* gene. Data are means $\pm$ SD of three biological replicates. Asterisks indicate a significant difference (*, $P < 0.05$; **, $P < 0.01$, *t*-test) compared with the corresponding control (0 h).

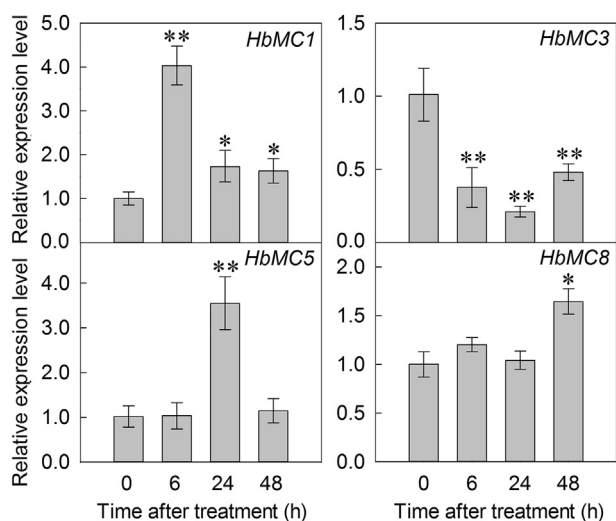


Fig. 9. Expression profiles of *HbMC* genes in latex of rubber tree responding to wounding. Latex was sampled at 0, 6, 24 and 48 h after wounding treatment. Relative expression level of each gene was normalized with *18S rRNA* gene. Data are means $\pm$ SD of three biological replicates. Asterisks indicate a significant difference (*, $P < 0.05$; **, $P < 0.01$, *t*-test) compared with the corresponding control (0 h).

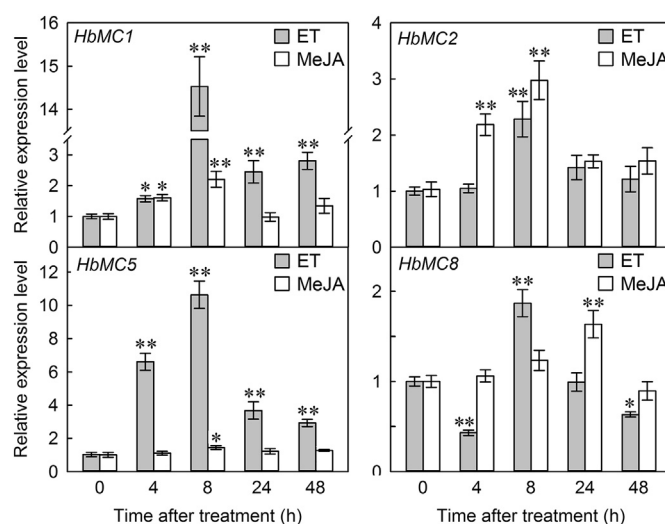


Fig. 10. Expression profiles of *HbMC* genes in latex of rubber tree responding to ET and MeJA treatments. Latex was sampled at 0, 4, 8, 24 and 48 h after treatments. Relative expression level of each gene was normalized with *18S rRNA* gene. Data are means $\pm$ SD of three biological replicates. Asterisks indicate a significant difference (*, $P < 0.05$; **, $P < 0.01$, *t*-test) compared with the corresponding control (0 h).

2015; Zhang et al., 2013). More recently, Huang et al. (2015) reported that members of the *OsMC* family displayed differential expression patterns in response to abiotic stress. In the present study, all of the *HbMCs*, except *HbMC5*, showed transcriptional changes when responding to cold, drought, and salt stresses (Figs. 6–8). Among them, *HbMC9* was strongly induced by cold and drought stresses, whereas its expression was not affected by salt stress. However, *OsMC7*, which is most closely related to *HbMC9* in

rice, showed significantly down-regulated expression in leaves after drought, cold, and salt stress treatments. Interestingly, *OsMC7*, showed opposite expression patterns in root after drought and salt stress treatments (Huang et al., 2015). The expression of *HbMC3* was significantly down-regulated by drought, cold, and salt stress. By contrast, *HbMC1* was significantly up-regulated by drought, cold, and salt stress (Figs. 6–8). *AtMC1*, the closest *Arabidopsis* homologue to *HbMC1*, was found to be a positive regulator of pathogen-

triggered PCD (Coll et al., 2014). Therefore, we speculate that *HbMC1* may positively regulate abiotic stress-induced PCD.

In the present study, *HbMC1*, *HbMC2*, *HbMC5*, and *HbMC8* showed significantly higher expression levels in TPD trees than healthy trees, suggesting the involvement of metacaspase-mediated PCD in TPD occurrence. Numerous studies have shown that TPD is a complex physiological disorder resulting from over tapping (wounding stress) or over stimulation by ethephon treatment (Faridah et al., 1996; Jacob et al., 1994; Putranto et al., 2015). Interestingly, we found that *HbMC1*, *HbMC5*, and *HbMC8* displayed up-regulation after wounding treatment (Fig. 9). In addition, they were all up-regulated by ET and MeJA treatments (Fig. 10). In some cases, ET and JA can induce ROS production and trigger PCD (Lam, 2004; Putranto et al., 2015; Zhang and Xing, 2008). Taken together, we speculate that *HbMC1*, *HbMC2*, *HbMC5*, and *HbMC8* may be involved in triggering PCD during the onset of TPD syndrome.

5. Conclusion

In this study, nine putative metacaspase genes (*HbMCs*) were identified in rubber tree. Subsequently, their bioinformatic characteristics were systematically analyzed, including conserved domains, sequence identities, exon-intron structures, and phylogenetic relationships. Expression profiling revealed that *HbMCs* were differently expressed in various tissues and regulated by abiotic stresses, implying that *HbMCs* may be involved in developmentally and/or environmentally regulated PCD. Additionally, *HbMC1*, *HbMC2*, *HbMC5*, and *HbMC8* were found to be associated with TPD. Our results present a comprehensive overview of rubber tree metacaspase gene family and lay an important foundation for further functional characterization of this family in *Hevea brasiliensis*.

Conflict of interest

The authors declare that they have no conflict of interest.

Author contributions

Hui Liu and Dejun Li designed the experiments and wrote the manuscript. Hui Liu, Zhi Deng, Jiangshu Chen, Sen Wang, and Lili Hao performed the experiments, data collection and statistical analysis. All of the authors read and approved the final manuscript.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.plaphy.2016.04.011>.

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